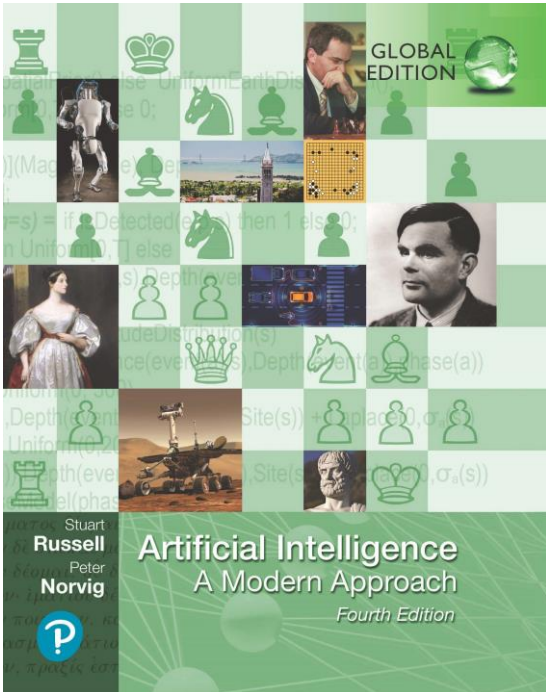


Chapter 4

Search with Uncertainty

Homework # 4



Implementation of the Erratic Vacuum World Intelligent Agent

- Write a Python program to implement the Erratic Vacuum World with Nondeterministic Actions Intelligent agent where search tree is defined using Python.
 - Program should be flexible to select any state as initial path to reach the final state.
 - Use the definitions on the next page.
- **Nondeterministic Actions** - Erratic Vacuum World
 - Regular fully-observable vacuum world, but the action '**suck**' is more powerful and **nondeterministic**:
 - a) **On a dirty square**: cleans the square and **sometimes cleans dirt on adjacent squares** as well
 - b) **On a clean square**: **sometimes deposits some dirt on the square**

